Flapping on Command: a *Drosophila* Escape Pathway as the Controller of a Flappy Bird Agent

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Abstract

Escape in flies is triggered by a compact pathway in which looming-sensitive visual neurons converge on a small set of descending neurons. That pathway is extracted here from the FlyWire adult brain reconstruction (384 neurons, 5810 connections), its weights are frozen, and it is used as the controller of a Flappy Bird agent. Only 64 interface numbers are optimised by an evolution strategy; each run converges in 8–11 generations and about 50 s on a laptop CPU. In all 5 runs the maximum of 100 pipes is cleared in every one of 200 unseen courses, whereas 84.2 pipes are cleared on average by a tuned hand-written rule. Silencing experiments show why the task is easier than it looks: the input neurons contact the output neurons directly, so behaviour is unaffected when all 60 intermediate neurons are removed, while individual runs come to depend on different single cell types. The study is a reproducible demonstration, not a model of escape behaviour.

Keywords: connectome; escape pathway; giant fiber; neuroevolution; closed-loop control

1 Introduction

The adult fly connectome released by the FlyWire consortium [1], together with a parts list and wiring diagram of the visual system [6] and neurotransmitter predictions obtained from the electron-microscopy images [2], makes small, behaviourally meaningful circuits easy to isolate. One of the best characterised is the looming-escape pathway. LPLC2 neurons were described as ultra-selective looming detectors whose selectivity arises from radial motion opponency [4]. In the giant fiber escape circuit, one visual input type was found to encode angular expansion velocity and another angular size, and motor program selection and timing were shown to emerge from the linear integration of these two features within the giant fiber [10].

Simulation of connectome circuits at the level of activity is becoming routine: a leaky integrate-and-fire model of the whole fly brain, built from connectivity and transmitter identity, predicted which neurons respond to tastes and which are required for feeding initiation [8]. A closed sensorimotor loop, in which the output of the circuit changes the stimuli it receives next, is rarely included in such models. That loop is closed here with the simplest possible one-button task. Flappy Bird is a natural match for an escape circuit, because the only decision is when to trigger an upward impulse. The game has also been used to compare agents obtained by evolutionary search and by deep reinforcement learning [9].

Methodologically, the network is a fixed recurrent circuit with trained input and output maps, an arrangement studied for generic recurrent circuits under the name of liquid state machines [5]. The interface is optimised with CMA-ES [3]. Evolution strategies of this kind have been proposed as

a scalable, easily parallelised alternative to reinforcement learning [7], which suits small controllers evaluated on CPUs.

Three things are provided: a minimal, fully seeded pipeline from the connectome tables to a playing agent; a quantitative report over independent runs and reference agents; and a silencing analysis that identifies which part of the circuit is actually used by the learned solutions.

2 Methods

Circuit. From the FlyWire v783 tables, LPLC2 (210 cells) and LC4 (104) were taken as inputs and the descending neurons DNp01 (the giant fiber), DNp02, DNp04, DNp06 and DNp11 (two each) as outputs. Any further neuron receiving at least 40 synapses in total from the inputs and sending at least 40 to the outputs was kept as an intermediate (60 cells). This gave 384 neurons and 5810 connections of at least five synapses. Weights were set to $\pm\log(1+\text{synapses})$ with signs from predicted transmitters [2]: 5264 cholinergic connections positive, 443 GABAergic and 103 glutamatergic negative, so that 90.6% of connections are excitatory. The weight matrix was scaled to spectral radius 0.9. Of the connections, 804 run directly from inputs to outputs, 2540 from inputs to intermediates and 215 from intermediates to outputs (Fig. 1).

Task. In a unit-height world the bird is placed at $x = 0.25$ with radius 0.025. Per frame, 0.002 is subtracted from the vertical velocity (fall speed capped at 0.04), and a flap sets the velocity to $+0.02$. Pipes are 0.10 wide, spaced 0.55 apart and moved left by 0.01 per frame; their gaps are 0.30 high, with centres drawn uniformly from $[0.25, 0.75]$ by a seeded xorshift generator. A run ends on contact with the floor, the ceiling or a pipe, and is won after 100 pipes. These constants were fixed before any network was trained, such that the reference rule below plays well.

Network and training. Six readings per frame (height, vertical velocity, horizontal distance to the next pipe, and the gap centre, top and bottom relative to the bird) are mapped onto eight input groups by a trained linear map. States follow $x \leftarrow x + a(-x + g\,W\tanh(x) + I)$ with four updates per frame. The mean tanh activity of each output cell type enters a trained weighted sum, and a flap is issued when the sum is positive. The trained parameters are the input map and biases, five readout weights, one bias, the leak and the gain, 64 in total. CMA-ES [3] was run with population 16 and eight courses per candidate until 1,000,000 frames had been simulated. Disjoint seeds were used for training, validation and test courses, and the procedure was repeated for 5 seeds.

References and silencing. Two reference agents were evaluated on the same test courses: a rule that flaps when the height predicted three frames ahead falls below the gap centre minus 0.08, and an agent that flaps in a random 10% of frames. For the lesion analysis, the tanh output of a group of neurons was forced to zero and the test courses were replayed.

3 Results

In every trained network, 100 of 100 pipes were cleared in all 200 test courses (mean 100.0, 100% wins), after 8–11 generations and at most 55 s of optimisation. On the same courses, 84.2 pipes were cleared by the hand-written rule (68% complete runs) and 0.07 by random flapping. The behaviour consists of a steady sequence of short flaps interleaved with glides, with longer climbs when consecutive gaps differ in height (Fig. 2).

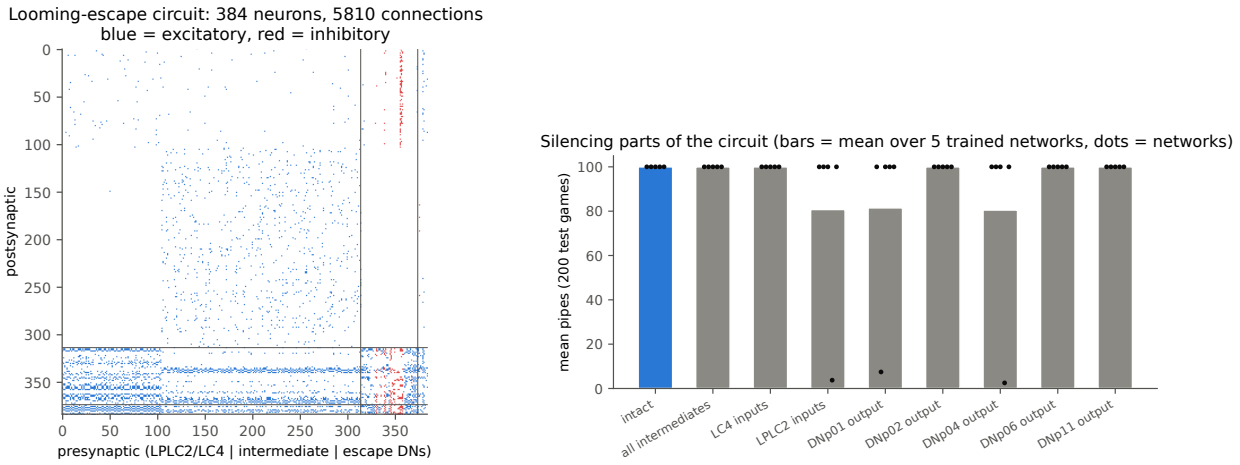


Figure 1: Left: the extracted escape pathway (inputs, intermediates, outputs; blue excitatory, red inhibitory). Right: mean pipes cleared after silencing a group of neurons; dots are the five trained networks.

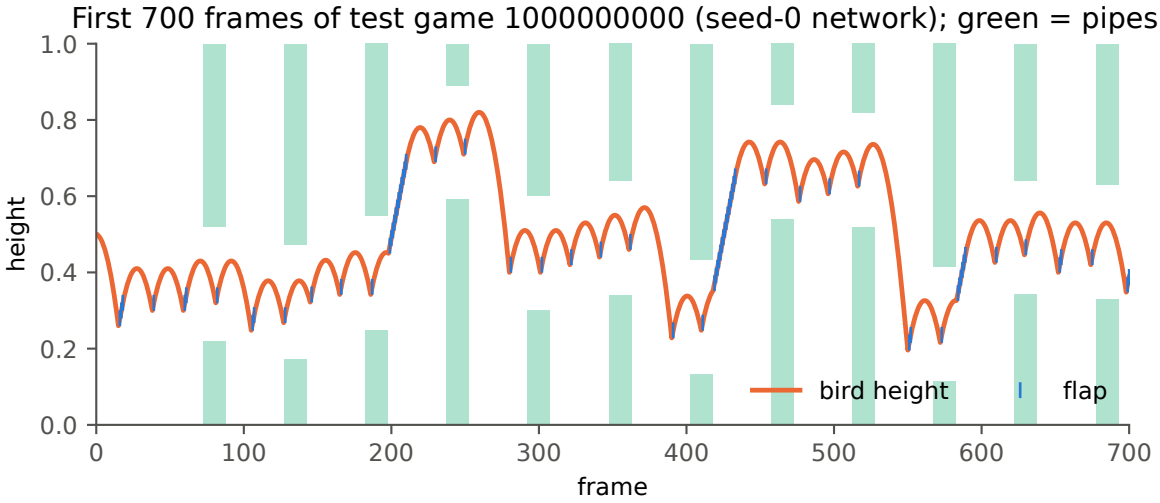


Figure 2: Height and flap times over the first 700 frames of a test course; green bands are pipes.

The ease of the task is explained by the silencing results. Nothing changed in any of the five networks when all 60 intermediate neurons were removed, which is possible because the inputs and outputs are bridged by 804 direct connections. Most single-group silencings were equally harmless: a network dropped below 50 pipes in only 3 of the 5×8 conditions. Those few cases are informative because they differ across runs: one network collapsed without the giant fiber DNp01, another without LPLC2 or without DNp04. Different routes through the same frozen circuit were therefore found by independent optimisations.

4 Discussion

A frozen escape pathway plus 64 trained numbers is sufficient to play this game perfectly, which is mainly a statement about the task: the gap is already localised by the six readings, and the game is played reasonably well by a two-condition rule. Matched random graphs were not evaluated in this

study, so no advantage of the measured wiring is claimed. The comparison that is reported, between parts of the circuit, shows that the short direct route is preferred by the learned solutions and that the loss of the multisynaptic route is tolerated. Biologically the model is deliberately thin: rate units are used instead of spiking neurons, synapse counts serve as weights, transmitter identities are predicted rather than measured, and neither optic-lobe preprocessing nor a body is included. Flies also do not flap through pipes; the analogy between triggering an escape and triggering a flap guided the choice of circuit and nothing more. A natural extension would be to drive the same pathway with rendered looming stimuli instead of abstract readings.

5 Conclusion

The looming-escape pathway of *Drosophila*, taken from FlyWire and left unchanged, controls a Flappy Bird agent by which every test course is completed after about a minute of CPU training.

Data and code availability. The connectome tables (FlyWire v783) are obtained from the FlyWire Codex portal [1]. Scripts, configuration files, raw logs and the JavaScript re-implementation used by the demo can be found at `https://github.com/burakgulmez/fly-flappybird`.

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